

3 Paley-Wiener theorem

In this section we change our point of view somewhat: we do not suppose any analyticity of f , but we do assume the validity of the Fourier inversion formula

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x \xi} d\xi \quad \text{if} \quad \hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi} dx,$$

under the conditions $|f(x)| \leq A/(1+x^2)$ and $|\hat{f}(\xi)| \leq A'/(1+\xi^2)$. For a proof of the inversion formula under these conditions, we refer the reader to Chapter 5 in Book I.

We start by pointing out a partial converse to Theorem 2.1.

Theorem 3.1 *Suppose $\hat{f}$ satisfies the decay condition $|\hat{f}(\xi)| \leq A e^{-2\pi a|\xi|}$ for some constants $a, A > 0$. Then $f(x)$ is the restriction to $\mathbb{R}$ of a function $f(z)$ holomorphic in the strip $S_b = \{z \in \mathbb{C} : |\operatorname{Im}(z)| < b\}$, for any $0 < b < a$.*

Proof. Define

$$f_n(z) = \int_{-n}^n \hat{f}(\xi) e^{2\pi i \xi z} d\xi,$$

and note that f_n is entire by Theorem 5.4 in Chapter 2. Observe also that $f(z)$ may be defined for all z in the strip S_b by

$$f(z) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i \xi z} d\xi,$$

because the integral converges absolutely by our assumption on $\hat{f}$: it is majorized by

$$A \int_{-\infty}^{\infty} e^{-2\pi a|\xi|} e^{2\pi b|\xi|} d\xi,$$

which is finite if $b < a$. Moreover, for $z \in S_b$

$$\begin{aligned} |f(z) - f_n(z)| &\leq A \int_{|\xi| \geq n} e^{-2\pi a|\xi|} e^{2\pi b|\xi|} d\xi \\ &\rightarrow 0 \quad \text{as } n \rightarrow \infty, \end{aligned}$$

and thus the sequence $\{f_n\}$ converges to f uniformly in S_b , which, by Theorem 5.2 in Chapter 2, proves the theorem.

We digress briefly to make the following observation.

Corollary 3.2 *If $\hat{f}(\xi) = O(e^{-2\pi a|\xi|})$ for some $a > 0$, and f vanishes in a non-empty open interval, then $f = 0$.*

Since by the theorem f is analytic in a region containing the real line, the corollary is a consequence of Theorem 4.8 in Chapter 2. In particular, we recover the fact proved in Exercise 21, Chapter 5 in Book I, namely that f and $\hat{f}$ cannot both have compact support unless $f = 0$.

The Paley-Wiener theorem goes a step further than the previous theorem, and describes the nature of those functions whose Fourier transforms are supported in a given interval $[-M, M]$.

Theorem 3.3 *Suppose f is continuous and of moderate decrease on $\mathbb{R}$. Then, f has an extension to the complex plane that is entire with $|f(z)| \leq Ae^{2\pi M|z|}$ for some $A > 0$, if and only if $\hat{f}$ is supported in the interval $[-M, M]$.*

One direction is simple. Suppose $\hat{f}$ is supported in $[-M, M]$. Then both f and $\hat{f}$ have moderate decrease, and the Fourier inversion formula applies

$$f(x) = \int_{-M}^M \hat{f}(\xi) e^{2\pi i \xi x} d\xi.$$

Since the range of integration is finite, we may replace x by the complex variable z in the integral, thereby defining a complex-valued function on $\mathbb{C}$ by

$$g(z) = \int_{-M}^M \hat{f}(\xi) e^{2\pi i \xi z} d\xi.$$

By construction $g(z) = f(z)$ if z is real, and g is holomorphic by Theorem 5.4 in Chapter 2. Finally, if $z = x + iy$, we have

$$\begin{aligned} |g(z)| &\leq \int_{-M}^M |\hat{f}(\xi)| e^{-2\pi \xi y} d\xi \\ &\leq Ae^{2\pi M|z|}. \end{aligned}$$

The converse result requires a little more work. It starts with the observation that if $\hat{f}$ were supported in $[-M, M]$, then the argument above would give the stronger bound $|f(z)| \leq Ae^{2\pi|y|}$ instead of what we assume, that is $|f(z)| \leq Ae^{2\pi|z|}$. The idea is then to try to reduce to the better situation, where this stronger bound holds. However, this is not quite enough because we need in addition a (moderate) decay as $x \rightarrow \infty$

(when $y \neq 0$) to deal with the convergence of certain integrals at infinity. Thus we begin by also assuming this further property of f , and then we remove the additional assumptions, one step at a time.

Step 1. We first assume that f is holomorphic in the complex plane, and satisfies the following condition regarding decay in x and growth in y :

$$(6) \quad |f(x + iy)| \leq A' \frac{e^{2\pi M|y|}}{1 + x^2}.$$

We then prove under this stronger assumption that $\hat{f}(\xi) = 0$ if $|\xi| > M$. To see this, we first suppose that $\xi > M$ and write

$$\begin{aligned} \hat{f}(\xi) &= \int_{-\infty}^{\infty} f(x) e^{-2\pi i \xi x} dx \\ &= \int_{-\infty}^{\infty} f(x - iy) e^{-2\pi i \xi (x - iy)} dx. \end{aligned}$$

Here we have shifted the real line down by an amount $y > 0$ using the standard argument (equation (1)). Putting absolute values gives the bound

$$\begin{aligned} |\hat{f}(\xi)| &\leq A' \int_{-\infty}^{\infty} \frac{e^{2\pi M y - 2\pi \xi y}}{1 + x^2} dx \\ &\leq C e^{-2\pi y(\xi - M)}. \end{aligned}$$

Letting y tend to infinity, and recalling that $\xi - M > 0$, proves that $\hat{f}(\xi) = 0$. A similar argument, shifting the contour up by $y > 0$, proves that $\hat{f}(\xi) = 0$ whenever $\xi < -M$.

Step 2. We relax condition (6) by assuming only that f satisfies

$$(7) \quad |f(x + iy)| \leq A e^{2\pi M|y|}.$$

This is still a stronger condition than in the theorem, but it is weaker than (6). Suppose first that $\xi > M$, and for $\epsilon > 0$ consider the following auxiliary function

$$f_{\epsilon}(z) = \frac{f(z)}{(1 + i\epsilon z)^2}.$$

We observe that the quantity $1/(1 + i\epsilon z)^2$ has absolute value less than or equal to 1 in the closed lower half-plane (including the real line) and

converges to 1 as ϵ tends to 0. In particular, this shows that $\widehat{f}_\epsilon(\xi) \rightarrow \widehat{f}(\xi)$ as $\epsilon \rightarrow 0$ since we may write

$$|\widehat{f}_\epsilon(\xi) - \widehat{f}(\xi)| \leq \int_{-\infty}^{\infty} |f(x)| \left[\frac{1}{(1 + i\epsilon x)^2} - 1 \right] dx,$$

and recall that f has moderate decrease on $\mathbb{R}$.

But for each fixed ϵ , we have

$$|f_\epsilon(x + iy)| \leq A'' \frac{e^{2\pi M|y|}}{1 + x^2},$$

so by Step 1 we must have $\widehat{f}_\epsilon(\xi) = 0$, and hence $\widehat{f}(\xi) = 0$ after passing to the limit as $\epsilon \rightarrow 0$. A similar argument applies if $\xi < -M$, although we must now argue in the upper half-plane, and use the factor $1/(1 - i\epsilon z)^2$ instead.

Step 3. To conclude the proof, it suffices to show that the conditions in the theorem imply condition (7) in Step 2. In fact, after dividing by an appropriate constant, it suffices to show that if $|f(x)| \leq 1$ for all real x , and $|f(z)| \leq e^{2\pi M|z|}$ for all complex z , then

$$|f(x + iy)| \leq e^{2\pi M|y|}.$$

This will follow from an ingenious and very useful idea of Phragmén and Lindelöf that allows one to adapt the maximum modulus principle to various unbounded regions. The particular result we need is as follows.

Theorem 3.4 *Suppose F is a holomorphic function in the sector*

$$S = \{z : -\pi/4 < \arg z < \pi/4\}$$

that is continuous on the closure of S . Assume $|F(z)| \leq 1$ on the boundary of the sector, and that there are constants $C, c > 0$ such that $|F(z)| \leq Ce^{c|z|}$ for all z in the sector. Then

$$|F(z)| \leq 1 \quad \text{for all } z \in S.$$

In other words, if F is bounded by 1 on the boundary of S and has no more than a reasonable amount of growth, then F is actually bounded everywhere by 1. That some restriction on the growth of F is necessary follows from a simple observation. Consider the function $F(z) = e^{z^2}$. Then F is bounded by 1 on the boundary of S , but if x is real, $F(x)$ is unbounded as $x \rightarrow \infty$. We now give the proof of Theorem 3.4.

Proof. The idea is to subdue the “enemy” function e^{z^2} and turn it to our advantage: in brief, one modifies e^{z^2} by replacing it by e^{z^α} with $\alpha < 2$. For simplicity we take the case $\alpha = 3/2$.

If $\epsilon > 0$, let

$$F_\epsilon(z) = F(z)e^{-\epsilon z^{3/2}}.$$

Here we have chosen the principal branch of the logarithm to define $z^{3/2}$ so that if $z = re^{i\theta}$ (with $-\pi < \theta < \pi$), then $z^{3/2} = r^{3/2}e^{3i\theta/2}$. Hence F_ϵ is holomorphic in S and continuous up to its boundary. Also

$$|e^{-\epsilon z^{3/2}}| = e^{-\epsilon r^{3/2} \cos(3\theta/2)};$$

and since $-\pi/4 < \theta < \pi/4$ in the sector, we get the inequalities

$$-\frac{\pi}{2} < -\frac{3\pi}{8} < \frac{3\theta}{2} < \frac{3\pi}{8} < \frac{\pi}{2},$$

and therefore $\cos(3\theta/2)$ is strictly positive in the sector. This, together with the fact that $|F(z)| \leq Ce^{c|z|}$, shows that $F_\epsilon(z)$ decreases rapidly in the closed sector as $|z| \rightarrow \infty$, and in particular F_ϵ is bounded. We claim that in fact $|F_\epsilon(z)| \leq 1$ for all $z \in \overline{S}$, where $\overline{S}$ denotes the closure of S . To prove this, we define

$$M = \sup_{z \in \overline{S}} |F_\epsilon(z)|.$$

Assuming F is not identically zero, let $\{w_j\}$ be a sequence of points such that $|F_\epsilon(w_j)| \rightarrow M$. Since $M \neq 0$ and F_ϵ decays to 0 as $|z|$ becomes large in the sector, w_j cannot escape to infinity, and we conclude that this sequence accumulates to a point $w \in \overline{S}$. By the maximum principle, w cannot be an interior point of S , so w lies on its boundary. But on the boundary, we have first $|F(z)| \leq 1$ by assumption, and also $|e^{-\epsilon z^{3/2}}| \leq 1$, so that $M \leq 1$, and the claim is proved.

Finally, we may let ϵ tend to 0 to conclude the proof of the theorem.

Further generalizations of the Phragmén-Lindelöf theorem are included in Exercise 9 and Problem 3.

We must now use this result to conclude the proof of the Paley-Wiener theorem, that is, show that if $|f(x)| \leq 1$ and $|f(z)| \leq e^{2\pi M|z|}$, then $|f(z)| \leq e^{2\pi M|y|}$. First, note that the sector in the Phragmén-Lindelöf theorem can be rotated, say to the first quadrant $Q = \{z = x + iy : x > 0, y > 0\}$, and the result remains the same. Then, we consider

$$F(z) = f(z)e^{2\pi i M z},$$

and note that F is bounded by 1 on the positive real and positive imaginary axes. Since we also have $|F(z)| \leq Ce^{c|z|}$ in the quadrant, we conclude by the Phragmén-Lindelöf theorem that $|F(z)| \leq 1$ for all z in Q , which implies $|f(z)| \leq e^{2\pi My}$. A similar argument for the other quadrants concludes Step 3 as well as the proof of the Paley-Wiener theorem.

We conclude with another version of the idea behind the Paley-Wiener theorem, this time characterizing the functions whose Fourier transform vanishes for all negative ξ .

Theorem 3.5 *Suppose f and $\hat{f}$ have moderate decrease. Then $\hat{f}(\xi) = 0$ for all $\xi < 0$ if and only if f can be extended to a continuous and bounded function in the closed upper half-plane $\{z = x + iy : y \geq 0\}$ with f holomorphic in the interior.*

Proof. First assume $\hat{f}(\xi) = 0$ for $\xi < 0$. By the Fourier inversion formula

$$f(x) = \int_0^\infty \hat{f}(\xi) e^{2\pi i x \xi} d\xi,$$

and we can extend f when $z = x + iy$ with $y \geq 0$ by

$$f(z) = \int_0^\infty \hat{f}(\xi) e^{2\pi i z \xi} d\xi.$$

Notice that the integral converges and that

$$|f(z)| \leq A \int_0^\infty \frac{d\xi}{1 + \xi^2} < \infty,$$

which proves the boundedness of f . The uniform convergence of

$$f_n(z) = \int_0^n \hat{f}(\xi) e^{2\pi i z \xi} d\xi$$

to $f(z)$ in the closed half-plane establishes the continuity of f there, and its holomorphicity in the interior.

For the converse, we argue in the spirit of the proof of Theorem 3.3. For ϵ and δ positive, we set

$$f_{\epsilon, \delta}(z) = \frac{f(z + i\delta)}{(1 - i\epsilon z)^2}.$$

Then $f_{\epsilon, \delta}$ is holomorphic in a region containing the closed upper half-plane. One also shows as before, using Cauchy's theorem, that $\widehat{f_{\epsilon, \delta}}(\xi) = 0$

for all $\xi < 0$. Then, passing to the limit successively, one has $\widehat{f_{\epsilon,0}}(\xi) = 0$ for $\xi < 0$, and finally $\hat{f}(\xi) = \widehat{f_{0,0}}(\xi) = 0$ for all $\xi < 0$.

Remark. The reader should note a certain analogy between the above theorem and Theorem 7.1 in Chapter 3. Here we are dealing with a function holomorphic in the upper half-plane, and there with a function holomorphic in a disc. In the present case the Fourier transform vanishes when $\xi < 0$, and in the earlier case, the Fourier coefficients vanish when $n < 0$.

4 Exercises

1. Suppose f is continuous and of moderate decrease, and $\hat{f}(\xi) = 0$ for all $\xi \in \mathbb{R}$. Show that $f = 0$ by completing the following outline:

(a) For each fixed real number t consider the two functions

$$A(z) = \int_{-\infty}^t f(x) e^{-2\pi i z(x-t)} dx \quad \text{and} \quad B(z) = - \int_t^{\infty} f(x) e^{-2\pi i z(x-t)} dx.$$

Show that $A(\xi) = B(\xi)$ for all $\xi \in \mathbb{R}$.

(b) Prove that the function F equal to A in the closed upper half-plane, and B in the lower half-plane, is entire and bounded, thus constant. In fact, show that $F = 0$.

(c) Deduce that

$$\int_{-\infty}^t f(x) dx = 0,$$

for all t , and conclude that $f = 0$.

2. If $f \in \mathfrak{F}_a$ with $a > 0$, then for any positive integer n one has $f^{(n)} \in \mathfrak{F}_b$ whenever $0 \leq b < a$.

[Hint: Modify the solution to Exercise 8 in Chapter 2.]

3. Show, by contour integration, that if $a > 0$ and $\xi \in \mathbb{R}$ then

$$\frac{1}{\pi} \int_{-\infty}^{\infty} \frac{a}{a^2 + x^2} e^{-2\pi i x \xi} dx = e^{-2\pi a |\xi|},$$

and check that

$$\int_{-\infty}^{\infty} e^{-2\pi a |\xi|} e^{2\pi i \xi x} d\xi = \frac{1}{\pi} \frac{a}{a^2 + x^2}.$$